

# Soumadeep Saha, PhD

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## Summary

My current research at ANITI explores techniques to better understand *reasoning with large language models*. I am also interested in alignment, and my recent work has received support from OpenAI. My PhD research centered on enhancing deep learning systems with the strengths of symbolic AI techniques for greater logical coherence and domain constraint adherence. While my primary focus is *natural language processing*, I actively collaborate across domains including vision, medicine, biology, astrophysics, and business applications.

## Education

Indian Statistical Institute, Kolkata

KOLKATA, INDIA

PhD (Computer Science)

2020–2025

☐ Supervisor: Prof. Utpal Garain

☐ Dissertation: “Domain Obedient Deep Learning”.

Indian Institute of Science Education and Research, Kolkata

KOLKATA, INDIA

Integrated BS & MS in Physics (Minor in Mathematics)

2015–2020

☐ Awarded the Kishore Vigyan Pratoshan Yojna (KVPY) fellowship from the Department of Science and Technology, Govt. of India (GoI).

☐ Master’s dissertation: “Towards Robust Deep Learning Systems”.

Bhavan’s G.K. Vidyamandir

KOLKATA, INDIA

10+2 (Pre-University Secondary Education)

2002–2015

☐ Recipient of the prestigious National Talent Search Examination (NTSE) Scholarship from NCERT, GoI.

☐ CBSE (10th standard): 91.2%; AISSCE (12th standard): 92%.

## Publications

☐ KisMATH: Do LLMs Have Knowledge of Implicit Structures in Mathematical Reasoning?

Accepted to TACL (12/01/2026)

S. Saha, A. Chaturvedi, S. Saha, U. Garain, N. Asher

In press.

☐ sudoLLM: On Multi-role Alignment of Language Models

EMNLP 2025 (findings)

S. Saha, A. Chaturvedi, J. Mahapatra, U. Garain

☐ Language Models are Crossword Solvers

NAACL 2025 (main)

S. Saha, S. Chakraborty, S. Saha, U. Garain

☐ Analyzing Semantic Faithfulness of Language Models via Input Intervention on Conversational Question Answering

Computational Linguistics (2024), 50(1): 119-155

A. Chaturvedi, S. Bhar, S. Saha, U. Garain, N. Asher

☐ On Measuring Intrinsic Causal Attributions in Deep Neural Networks

CLear 2025

S. Saha, D. V. Rathore, S. Saha, D. Doermann, U. Garain.

☐ VALUED - Vision and Logical Understanding Evaluation Dataset

DMLR (2024), (13):1-18.

S. Saha, S. Saha, U. Garain

☐ MedTric : A clinically applicable metric for evaluation of multi-label computational diagnostic systems

PLOS One (2023), 18(8): e0283895

S. Saha, U. Garain, A. Ukil, A. Pal, S. Khandelwal.

☐ DOST – Domain Obedient Self-supervised Training for Multi Label Classification with Noisy Labels

AAAI 2024 Workshop (W3PHIAI) (2024), 1164: 117-127

S. Saha, U. Garain, A. Ukil, A. Pal, S. Khandelwal.

☐ LADDER: Revisiting the Cosmic Distance Ladder with Deep Learning Approaches and Exploring its Applications

ApJS (2024), 273(2): 27

R. Shah, S. Saha, P. Mukherjee, U. Garain, S. Pal

☐ The Synergistic Integration of Access Control Management and Large Language Model Agents: A Survey

Preprint. 10.36227/techrxiv.177075184.43798334/v1

F. Periti, S. Saha

## Patents

❑ Method and System for Contradiction Avoided Learning for Multi-Class Multi-Label Classification

S. Saha, U. Garain, A. Ukil, A. Pal

US Patent - US12038949B2

Granted Jul. 2024

❑ Method and System for Evaluating Clinical Efficiency of Multi-Label Multi-Class Computational Diagnostic Models

U. Garain, S. Saha, A. Ukil, T. Deb, S. Richa A. Pal, S. Khandelwal

Pending

Filed Sep. 2022.

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## Experience

### Helmholtz Visiting Researcher

Jul '24–Sep '24

Awarded the Helmholtz Information and Data Science Academy (HIDA) visiting researcher grant to work at the Institute of Aerospace Medicine, DLR (German Aerospace Center). Conducted research on improving AI-based retinal diagnostics for spaceflight applications.

### Alleima

Mar '24–May '24

Developed a custom end-to-end computer vision-based solution for assembly-line object detection/tracking, enabling autonomous logging of crucial data with existing infrastructure.

### TCS Research

Nov '21 – Jul '22

Conducted research as part of a team on ECG-based diagnosis of cardiovascular diseases. Independently identified key gaps in the literature and developed innovative, state-of-the-art solutions to address them, resulting in two patents, and publications.

### Talks and Presentations

- ❑ Organized a 4-day workshop for DataLab, **Capital One**, Bangalore (2023).
- ❑ Subject matter expert for Be10x—designed MOOCs, and delivered live seminars to 500+ participants (2024).
- ❑ TA for Natural Language Processing course at ISI Kolkata (2022–2025).
- ❑ Instructor at the Winter School of Deep Learning (WSDL), ISI Kolkata (2021–2024).
- ❑ Instructor for the Comprehensive Course on Business Analytics, ISI Kolkata (2022).
- ❑ Presented my work on Logically Coherent Deep Learning at Amazon Research Days (ARD '22).
- ❑ Presented my work on domain-obedient self-supervision at W3PHIAI, AAAI 2024, Vancouver, Canada.
- ❑ Presented my work on constrained LLM inference at NAACL 2025, Albuquerque, USA.

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## Skills

- ❑ **Deep Learning:** Expert in deep learning theory and practice, from low-level optimization to advanced architectures and training paradigms.
- ❑ **Python:** 10 years of experience. Proficient in **pytorch**, **Hugging Face**, **numpy**, etc., with extensive experience developing custom modules, training loops, etc., and scaling for high-performance GPU clusters.
- ❑ **NLP:** Significant LLM expertise (in-context learning, RAG, PEFT, Q-LoRA, instruction tuning, RLHF, DPO, etc.)
- ❑ **Deep RL:** DQN, PPO, MCTS, GRPO, etc.
- ❑ **Programming:** In addition to python, experienced in C++, FORTRAN, bash, etc.
- ❑ **Vision:** Vision transformers, CNNs, VLMs, object detection, segmentation, etc.

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## Interests

- ❑ **Robotics/DIY**—Active interest in 3D printing, robotics, electronics, etc. I have conducted introductory workshops on robotics and was the Secretary of the Robotics and Astronomy Club at IISER Kolkata.
  - ❑ **Sports**—Represented my college in national-level sports meets in basketball and volleyball. Played in my state's Senior Division Men's Basketball League.
  - ❑ **Music**—Classically trained pianist.
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